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Galium verum L. extract mitigates cardiovascular events in psoriasis rats

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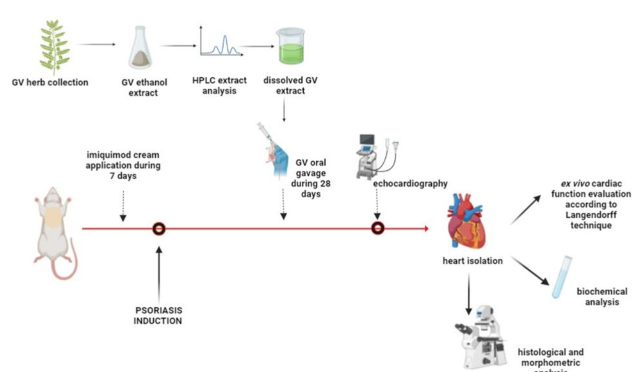
Abstract

Psoriasis has been considered a systemic immune-mediated disease that can affect function of the heart. However, certain herbal therapy approaches may nullify side effects of psoriasis on the heart. The aim of this study was to investigate the cardioprotective effects of *Galium verum* extract administration of the heart of psoriatic rats. The study included 24 *Wistar* albino male rats, divided into 4 groups: control (CTRL), *G.verum* (GV), psoriasis (PSORI), and psoriasis with *G.verum* (PSORI + GV). Seven-day topical application of 5% *imiquimod* cream was used for induction of psoriasis. After induction, animals were received 125 mg/kg *G.verum* extract for 4 weeks. Isolated hearts were perfused on the Langendorff apparatus and measured: dp/dt max/min, SLVP, DLVP, HR and coronary flow. The oxidative stress biomarkers: TBARS, NO2, O2- and H2O2 were measured in coronary venous effluent. Isolated hearts were fixed and stained with H/E and Picro-sirius red staining. Psoriasis decreased cardiac contractility and relaxation and increased SLVP and DLVP at all perfusion pressure. Treatment with *G.verum* in psoriasis rats improved contractility and relaxation of the heart and rise SLVP and DLVP. In PSORI + GV group, the decrease of oxidative stress biomarkers were observed in comparison to PSORI group. Diameter and cross-section area of cardiomyocytes were increased in PSORI and PSORI + GV groups compared to the control. Collagen content was increased in PSORI group by 283% and in PSORI + GV group by 188% compared to control. Treatment with *G.verum* extract exhibited a positive effect on cardiac function, morphometry and redox state of heart of psoriatic rats.

Graphic abstract

Flow chart of the experimental design. After preparing, the analysis of the obtained *G. verum* extract was carried out. Psoriasis was induced by applying 5% imiquimod cream to the shaved backs of rats for 7 days. After disease induction, animals were treated with *G. verum* extract for 28 days. After the end of the experimental protocol, echocardiographic evaluation, biochemical analysis, examination of cardiodinamic and morphometric parameters of the left ventricle were performed.



Keywords *Galium verum* · Psoriasis · Rat · Heart · Cardiodynamic · Morphometry

Extended author information available on the last page of the article

Introduction

Psoriasis is a chronic inflammatory disease, associated with stress, recurrent evolution and consequences for the patient's mental and physical health. It is assumed that 2–5% of the world's population suffers from psoriasis [1], while in Europe the incidence of the disease is between 0.6–6.5% and is especially prevalent in Northern European countries [2]. It is characterized by excessive proliferation of keratinocytes, scaly plaques, severe inflammation and erythema. These inflammatory changes are localized on the skin and joints, mostly on elbows and knees [1]. The mechanism that initiates the inflammatory cascade associated with psoriasis still remains unknown, but recent evidence suggests the role of the IL-23/IL-17A axis in the etiology of psoriasis [3]. The interaction between epidermal keratinocytes, leukocytes, T lymphocytes and vascular endothelium is involved in the pathophysiological mechanisms of psoriasis [4]. Besides increased leukocyte migration and increased levels of growth factors and cytokines, some genetic factors such as IL-1β, IL-6, IL-17, IL-22, IL-23, STAT-3, StAR, are the most critical factors, which affect the exacerbation of psoriatic lesions [4].

More than 50 years ago, autopsy in psoriasis patients has been discovered increased myocardial fibrosis and coronary atherosclerosis [5], together with damaged vascular endothelium in both lesional and non-lesional skin [6]. Psoriasis is considered as a systemic entity [7], because it is often associated with cardiometabolic disease, psoriatic arthritis [7] and hormonal changes [8]. Cardiovascular conditions significantly related with psoriasis are ischemic heart disease, angina and myocardial infarction, transient ischemic attack, stroke and peripheral vascular diseases. Cardiovascular conditions such as arrhythmia, structural heart disorders and pulmonary hypertension are sporadically related with psoriasis [9]. Metabolic syndrome (characterized by increased body mass index, hypertension, coronary artery disease, hyperlipidemia and diabetes type 2, as well as the presence of coronary plaques) are twofold frequent in psoriasis patients [7]. On the other hand, psoriasis patients with severe cutaneous lesions have up to a fivefold higher risk of developing dyslipidemia [10] as well as developing cardiovascular disease [11]. Components of the immune response in psoriasis such as increased T-cell and myeloid cell activation, platelet activation, tumor necrosis factor-α, IL-6, IL-17 and IL-23 are linked to the development of atherosclerosis and vascular inflammation [11, 12]. Psoriasis is an independent risk factor for increased osteopontin levels, an inflammatory glycoprotein that is important in the development of atherosclerosis [13]. Some literature data suggest that psoriasis skin disease treatment may lessen coronary plaque burden, vascular inflammation, and cardiovascular risk [11].

Current conventional therapy for the treatment of psoriasis, include the use of methotrexate, cyclosporine A, acitretin, as well as biological therapy –such as secukinumab or ustekinumab. It is known that conventional therapy has numerous side effects, such as skin atrophy, sensitivity to sunlight, irritation, high risk for infection, carcinogenesis, suppression of the immune system and toxic effects on various organ systems [2]. Due to the mentioned side effects and high treatment costs, the scientists are searching for a new therapeutic approach in psoriasis treatment that will primarily have less side effects. Herbal extracts have the central role in this novel approach. The advantages of such therapy are numerous and related to, fewer side effects and toxicity, as well as multiple biochemical activities. Literature data suggest that many patients with psoriasis are willing to use complementary and alternative medicine as one of the modalities for the treatment [14, 15]. Nowadays, about 43–69% of patients with psoriasis use different modalities of alternative medicine [14, 15], dominantly herbal preparations. It is known that some plant extracts have anti-inflammatory, anti-angiogenic, antioxidant and antimicrobial effects, and hence their justification in the attempt to create new medicines or supplements for the treatment of various diseases. Knowing the numerous benefits of using natural products in the treatment of various diseases, researchers are looking for new herbal formulas that, in addition to reducing the disease symptoms, will also improve the quality of life of patients [16].

Galium verum L. is known as a plant with multiple therapeutic properties and is widely used in traditional medicine, at the beginning as a choleretic, diuretic or spasmolytic treatment [17] and then as well as in the topical treatment of psoriasis [18] and wound healing [19]. Plant extracts contains bioactive compounds such as iridoid glycosides, flavonoids, terpenes, saponines, anthraquinones, phenolic acids, polysaccharide complex, aldehydes, alcohols, quercetin, small amounts of tannins, waxes, pigments, essential oils, luteolin and vitamin C, which are associated with the effectiveness of this plant [17–23]. The dried parts of the plant have been traditionally used for topical treatment of psoriasis [18]. The antioxidant, antimicrobial, cytotoxic and endocrine potential of *G. verum* were proven in previous studies [17, 18]. Evidence from the literature showed that oral administration of *G. verum* improves cardiodynamic parameters in the isolated rat heart [17]. However, there is no scientific evidence that this plant is effective in systemic treatment of psoriasis.

Bearing in mind all previously mentioned, the main goal of the performed investigation was to initially assess the hypothesis that psoriasis might induce alteration in cardiovascular functioning and is the treatment *G. verum* extract can improve cardiovascular health and local skin changes of psoriatic rats.

Materials and methods

Plant material and extract

The plant *G. verum* was collected in 2023 in the village of Dobroselica, Mt. Zlatibor (GPS coordinates: 43°42′59.99″ N, 19°41′59.99″ E). The identification and classification were carried out at the Institute of Botany and Botanical Garden "Jevremovac" at the University of Belgrade. The collected plant material was dried in the shade and ground using a 0.75 mm sieve. In order to prepare extract, 100 g of the aerial parts were extracted through heat reflux with 500 ml of 70% EtOH. The mixture was filtered using Whatman No. 1 filter paper, and the dry extract was obtained by evaporating the solvent under reduced pressure (RV05 basic IKA, Germany). The remaining residue was stored in a dark glass bottle at + 4 °C until needed. Each day, just before administration to the experimental animals, the extract was dissolved in tap water.

Chemical characterization of the G. verum extract

3,5-Di-tert-4-butylhydroxytoluene (BHT) analytical standard (≥ 99.0%, GC), 2,2-diphenyl-1-picrylhydrazyl radical (DPPH), and quercetin (≥ 95.0%, HPLC), procyanidin B1 and procyanidin B2 (≥ 90.0%, HPLC), p-hydroxzbenzoic acid (purity 99.0%), p-coumaric acid (purity 99.0%), protocatechuic acid (purity 99.0%), sinapic acid (purity 98.0%), and syringic acid (purity 95.0%) were purchased from Sigma (St Louis, MO, USA), PEG 400 from BASF, Germany. Gallic and chlorogenic acid (≥ 99.0%, HPLC), cyanidin chloride, pelargonidin chloride, peonidin chloride, delphinidin chloride, malvidin chloride (≥ 97.0%, HPLC), (-)-epicatechin and (+)-catechin, hyperoside, rutin, isoquercitrin (all, ≥ 99.0%, HPLC), were purchased from Extrasynthese (Genay Cedex, France). All other used chemicals were purchased from Merck (Germany).

Determination of total phenolic content

The total phenolic content was determined by the Folin-Ciocalteu method. The absorbance was measured by UV–VIS Spectrophotometer HP 8453 (Agilent Technologies, USA), at 725 nm. Gallic acid (0–100 mg/L) was used for calibration of a standard curve. The calibration curve showed the linear regression at $r > 0.99$, and the obtained results are expressed as milligrams of gallic acid equivalents per gram of juice or dry weight (mg GAE/g DW). Triplicate measurements were taken and data were presented as mean ± standard deviation (SD) [24].

Determination of total flavonoids content

The percentage content of flavonoids expressed as hyperoside, was calculated using the method described in DAB 10. Briefly, the sample was extracted with acetone/HCl under reflux condenser; the $AlCl_3$ complex of flavonoid fraction extracted by ethyl acetate was measured by UV–VIS Spectrophotometer HP 8453 at 425 nm. The content of flavonoid, expressed as hyperoside percentage, presented the mean ± standard deviation of three determinations was calculated using following expression: $A \times 1.25/m$, where A was absorbance at 425 nm and (m) was mass of the extracts to be examined in grams [25].

Determination of tannins content

The percentage content of tannins was calculated using the method described in the European Pharmacopoeia, Ph. Eur. 6.0. [26]. Shortly, decoctions, prepared from the investigated extracts, are treated with phosphomolybdotungstic reagent in alkaline medium after and without treatment with hide powder. The absorbance was measured by UV–VIS Spectrophotometer HP 8453 (Agilent Technologies, USA), at λmax 760 nm. From the difference in absorbance of total polyphenols and polyphenols not adsorbed by hide powder, the percentage content of tannins expressed as pyrogallol (%, w/w), were calculated from the expression:

$$62.5(A_1 - A_2) \times m_2$$

$$A_3 \times m_1$$

where m1 represents mass of the sample to be examined, in grams; and m2 is mass of pyrogallol, in grams. The results represent the mean ± SD of three determinations.

HPLC fingerprint

HPLC fingerprint of the investigated sample was achieved by HPLC (Agilent Technologies 1200). Detection was performed using Diode Array Detector (DAD), and the chromatograms were recorded at λ = 360 nm. HPLC separation of components was achieved using a LiChrospher 100 RP 18e (5 μm), 250 × 4 mm i.d. column, with a flow rate of 0.8 mL/min and mobile phase, A [500 mL of H2O plus 9.8 mL of 85% H3PO4 (w/w)], B (MeCN), elution, combination of gradient mode: 89–75% A, 0–35 min; 75–60% A, 35–55 min; 60–35% A, 55–60 min; 35–0% A, 60–70 min). A portion of the sample solution (12.22 mg/mL 70% EtOH, DW = 9.0%), previously prepared as described, was filtered through 0.2 μm PTFE filters (Fisher, Pittsburgh, PA) prior to HPLC analysis. The injected volume was 4 μL. Standard solutions for the determination of polyphenolic compounds were prepared in at a final concentration of 0.01 mg/mL (protocatechuic, p-coumaric), 0.02 mg/mL (chlorogenic acid, rutin), 0.05 mg/mL (quercetin). A phenolic standard library was constructed with each authentic standard at the above-mentioned concentrations, injected separately, and the data were acquired by the photodiode array detector. The volume injected was 4 µL, the same as the investigated extract. The identification was carried out thanks to retention time and spectra matching. Once spectra matching succeeded, results were confirmed by spiking with respective standards to achieve a complete identification by means of the so-called peak purity test, meaning that each peak was tested for purity by a three-point purity test and for the similarity by a library search comparing the peak spectrum to that of the standards. High similarity index and a common retention time with the standard were considered a positive identification. Those peaks not fulfilling these requirements were not quantified). Under the conditions employed in this study, the relative standard deviation for the retention times in three repetitive runs was in the range of 0.18—1.79%. Quantification was performed by external calibration with corresponding standards. The results represent the mean ± SD of three determinations.

Experimental animals

This study included 24 Wistar albino male rats (10 weeks old, weighing 200–250 g). The rats were obtained from the Military Medical Academy, Belgrade, Serbia. Rats were housed in plexiglass transparent cages (one rat per cage). The room temperature was kept at 22 ± 1 °C with 12:12 h light and dark cycles. Food and water were provided ad libitum.

The rats were randomly classified into 4 groups (6 animals per group):

1. CTRL - control group – untreated rats
2. GV - healthy rats with systemic administration of *G. verum* extract for 4 weeks
3. PSORI - psoriasis
4. PSORI + GV - psoriasis with systemic administration of *G. verum* extract for 4 weeks

Induction of psoriasis

The rats from psoriasis groups was treated by daily topical application of 50 mg 5% imiquimod cream (Aldara) on their shaved back skin (3 cm x 2.5 cm) for eight consecutive days. After induction, the extract of the investigated plant was administered daily by oral gavage. The beneficial dose of 125 mg/kg body weight [17] of the extract was administered to rats from the GV positive groups, during a period of four weeks. Psoriasis area severity index (PASI score), a modified version of human PASI score, was calculated after the induction of psoriasis and after one and two weeks of treatment. On this occasion, the following parameters was monitored: erythema, scaling and thickness of back skin and scored on a scale from 0–4 for each parameter (0—none; 1-slight; 2- moderate; 3—marked; and 4—very marked) [27], so the maximum PASI score will be 12.

Echocardiographic evaluation

After four-week administration of *G.verum* extract, transthoracic echocardiography evaluation was performed using a Hewlett-Packard Sonas 5500, (Andover, MA, USA) sector scanner. Rats were anesthetized with an intraperitoneal injection of a combination of ketamine (Ketamine 10%, CP-PHARMA, Burgdof, Germany; 100 mg/kg) and xylazine (Xyla, Interchemie, Holland; 10 mg/kg). By using M-mode following parameters were measured: interventricular septal wall thickness at end-diastole (IVSd), left ventricle (LV) internal dimension at end-diastole (LVIDd), LV posterior wall thickness at end-diastole (LVPWd), the interventricular septal wall thickness at end-systole (IVSs), LV internal dimension at end-systole (LVIDs), LV posterior wall thickness at end-systole (LVPWs), as well as fractional shortening percentage (FS%) [28].

Isolation of heart

After finished echocardiographic examination in short-term anesthesia, the animals were premedicated with heparin as an anticoagulant and sacrificed by cervical dislocation (Schedule 1 of the Animals/Scientific Procedures, Act 1986 UK). After an emergency thoracotomy, the hearts were quickly removed, the aortas were canulated, and retrogradely perfused according to modified Langerdorf techniques at a gradually raised coronary perfusion pressure (CPP) (40–120 H_2O). The Krebs–Henseleit perfusate had the following constituents: Mm/L: NaCl 118, KCl 4.7, $CaCl_2 \cdot 2H_2O$ 2.5, $MgSO_4 \cdot 7H_2O$ 1.7, $NaHCO_3$ 25 and KH_2PO_4 1.2, equilibrate with 95% O_2 and 5% CO_2 and warmed to 37 °C (pH 7.4). The sensor (transducer BS4 73–0184, Experimetria Ltd., Budapest, Hungary), used to continuously monitor cardiac function, was placed via the newly damaged left atrium and mitral valve into the left ventricle. The left ventricle was the location where the sensor was placed, and it continually recorded the following myocardial function parameters (cardiodinamic parameters): maximum rate of pressure development in the left ventricle (dp/dt max); minimum rate of pressure development in the left ventricle (dp/dt min); systolic left ventricular pressure (SLVP); diastolic left ventricular pressure (DLVP); heart rate (HR). Coronary flow (CF) was measured flowmetrically [29]. All research procedures were carried out in accordance with the European Council Directive (86/609/EEC) as well as the principles of Good Laboratory Practice (2004/9/EC, 2004/10/EC) and were approved by the Ethics Committee for the Welfare of Experimental Animals, Faculty of Medical Sciences, University of Kragujevac, Serbia (No. 01–1094815, from October 10, 2022).

Oxidative stress analysis

Following the stabilization phase in the ex vivo heart perfusion, coronary venous effluent samples were taken at each perfusion pressure in order to measure the oxidative stress markers such as index of lipid peroxidation (measured as TBARS), nitrites (NO_2), superoxide anion radical (O_2-), and hydrogen peroxide (H_2O_2).

Index of lipid peroxidation (TBARS)

The venous effluent's degree of lipid peroxidation was determined by measuring TBARS. 0.8 ml of sample and 0.4 ml of trichloroacetic acid were combined to complete the procedure. Following a 15 min incubation period at 100 °C, the samples was measured at wavelength 530 nm. The use of Krebs–Henseleit solution served as a blind control [30].

Nitrites (NO_{2}_)

Nitrite (NO_{2}_) was determined as an index of nitric oxide production with Griess reagent. 0.5 ml of the perfusate was precipitated with 200 µl of 30% sulfosalicylic acid, mixed for 30 min, and centrifuged at 3000 × g. Equal volumes of the supernatant and Griess reagent were mixed and stabilized for 10 min in the dark, and then, the sample was measured spectrophotometrically at a wavelength of 543 nm. Nitrite concentrations were determined using sodium nitrite as the standard [17].

Superoxide anion radical O_{2-}

The level of the O_{2-} was measured via a nitro blue tetrazolium (NBT) reaction in TRIS buffer with coronary venous effluent, at 530 nm. The use of Krebs–Henseleit solution served as a blind control [17].

Hydrogen peroxide (H_2O_2)

The measurement of the level of H_2O_2 was based on the oxidation of phenol red by H_2O_2 in a reaction catalyzed by horseradish peroxidase (HRPO) [17]. 200 µl of samples was precipitated using 800 µl of freshly prepared phenol red solution; 10 µl of (1:20) HRPO (made extempore) were subsequently added. As a blind probe, Krebs–Henseleit solution was used. The samples was measured at wavelength at 610 nm.

Tissue processing, histological staining and morphometric analysis

The isolated rat hearts were measured and halved so the left ventricle wall was fully exposed. The samples were fixed in 4% neutral formalin, dehydrated in increase concentration of ethanol (70%, 96%, and 100%), cleared in xylene and embedded in Histowax® (Histolab Product AB, Göteborg, Sweden), and processed for further histological analysis. 5 µm thick sections, were stained with Hematoxylin/Eosin (H/E) for detection of morphological changes and morphometric analysis. Picrosirius red staining was used for the detection of the collagen content. This method results in reddish stain of collagen, dark brown to black stained nuclei and yellodish muscle cell cytoplasm. The procedure included deparaffinization and rehydration (100%, 96% and 70% ethanol) of the tissue sections. After rehydration, sections were incubated in Weigert's hematoxylin solution for 5 min and washed in distilled water for 10 min. After washing, the sections were incubated in Picro-sirius red solution for 60 min. This was followed by rinsing in distilled water for 10 min, then incubated in a 2% solution of glacial acetic acid (2 × 5 min). Tissue sections were washed in distilled water, dehydrated in ethanol, cleared in xylene and mounted in DPX. The images of tissue sections were captured with a digital camera attached to the Olympus BX51 microscope. Morphometric analysis (cross-section area and diameter of the cardiomyocyte) was carried out using calibrated Axiovision software (Zeiss, White Plains, NY, USA) as well as with Image Pro-Plus (Media Cybernetics, Rockville, MD, USA) according to our previously describe methodology [29].

Statistical analysis

Data distribution was established using the Shapiro–Wilk test. Statistical calculations were performed with the SPSS computer program, version 20.0 (SPSS Inc., Chicago, IL, USA). Statistical comparison of data was used by one-way Anova test with the post hoc LSD test analysis for multiple comparisons. p values below 0.05 were considered statistically significant. All data are presented as mean values ± standard deviation (SD).

Results

Chemical analysis of G. verum extract

The results of chemical characterization revealed that TPC of *G. verum* ethanolic extract was 147.00 ± 3.21 mg GA/g d.w. extract, whereas TFC was 2.28 ± 0.10% (expressed as hyperoside percentage). Additionally, TTC was 3.26 ± 0.11% expressed as pyrogallol, w/w.

The chemical composition of the *G. verum* ethanolic extract is shown in Table 1 and Fig. 1.

The results indicate that the most abundant phenolic compound was rutin, followed by chlorogenic acid. Furthermore, significant abundance of luteolin-7-O-glucoside was recorded as well, while other phenolic compounds were detected in lower quantities (Table 1).

Effects of the treatment of G. verum extract on the skin of psoriatic rats

After induction of psoriasis, on the rats back skin was observed marked erythema and desquamation (Fig. 2a). After 7 days of treatment with *G. verum* extract, rats skin was noticeably less redness and desquamated (Fig. 2b). After 14 days of treatment, the redness of the skin has completely subsided. Only discrete desquamation remain present (Fig. 2c). The PASI score after induction was 9, after seven days it was 5 and after 14 days it was 1. By histological examination of rats skin, in the control group, there were no morphological changes in the structure of the skin (Fig. 2d). In the psoriasis group, we observed desquamation, hyperkeratosis and inflammation of rat skin (Fig. 2e). In the group of psoriatic rats on treatment with *G. verum* extract, the reduction of hyperkeratosis was observed as well as desquamation (Fig. 2f).

Table 1 Concentration of compounds identified in *G. verum* ethanolic extract (d.w. – dry weight)

Peak Number	Compound	Concentration (mg/g d.w. extract)
1	Protocatechuic acid[1]	0.48 ± 0.03
2	Chlorogenic acid[1]	20.93 ± 0.15
3	p-coumaric acid[1]	0.72 ± 0.05
4	Vitexin[2]	1.55 ± 0.09
5	Rutin[2]	25.0 ± 0.23
6	Luteolin-7-O-glucoside[2]	14.14 ± 0.13
7	3,5-di-O-caffeoylquinic acid[1]	5.1 ± 0.10
8	Quercitrin[2]	2.26 ± 0.07
9	Rosmarinic acid[1]	2.32 ± 0.07
10	Diosmin[2]	1.59 ± 0.03
11	Quercetin[2]	2.76 ± 0.06
12	Luteolin[2]	1.2 ± 0.03
13	Apigenin[2]	0.13 ± 0.02
14	Chrysoeriol[2]	0.25 ± 0.03
15	Isorhamnetin[2]	0.77 ± 0.05

Results are expressed as Mean ± Standard deviation

[1]Phenolic acid group of compounds

[2]Flavonoid group of compounds

Body weight (BW), heart weight (HW), and index hypertrophy of heart (HW/BW ratio)

The increase in body weight was observed in the PSORI and PSORI + GV groups by 16% and 17% (respectively) compared to control values. Heart weight was increased in both PSORI and PSORI + GV groups compared to the CTRL and GV groups. The index of hypertrophy of the heart (HW/BW ratio) was increased in the PSORI group by 14% and in the PSORI + GV group by 8.6% compared to control values. Treatment with *G. verum* in psoriatic rats decreased HW/BW ratio by 7% compared to the PSORI group (Table 2).

Echocardiographic evaluation

By echocardiographic examination, we noticed a statistically significant increase in IVSd, LVIDd, LVPWd, IVSs, LVIDs and decreased FS in the PSORI group compared to CTRL and GV groups. Treatment with *G. verum* extract in psoriatic rats significantly decreased IVSs, LVIDs, LVPWs and increased FS compared to the PSORI group (Table 3).

Cardiodinamic parameters

The values of the maximum rate of pressure development in the left ventricle (dp/dt max) in the PSORI and PSORI + GV groups were lower, while in the GV group the values were increased (statistically insignificantly) at all perfusion pressure compared to the control group (Fig. 3a). The increase in dp/dt min was observed in all experimental groups in all perfusion pressure compared to the control. Significant increase in dp/dt min at coronary perfusion pressure starting from 80 to 120 H_2O was observed in the PSORI group compared to the CTRL. Opposite to this, treatment with *G. verum* extract in psoriatic rats, decrease dp/dt min at coronary perfusion pressure starting from 80 to 120 H_2O compared to psoriasis alone (Fig. 3b). The values of systolic left ventricle pressure were increased in both psoriatic groups (PSORI and PSORI + GV) in all perfusion pressure compared to the control values. SLVP did not change in the GV group compared to CTRL. Treatment with *G. verum* in psoriatic rats decreased SLVP compared to the PSORI group (Fig. 3c). Diastolic left ventricle pressure was increased in the PSORI and PSORI + GV groups compared to control values. On the other hand, values of DLVP were significantly decreased in the GV group at coronary perfusion pressure 100 and 120 H_2O compared to control (Fig. 3d). The values of the heart rate were statistically significantly decreased in all experimental groups compared to control. The largest decrease in HR at all coronary perfusion pressure was observed in the PSORI group compared to CTRL (Fig. 3e). Coronary flow was increased in all experimental groups at all perfusion pressures compared to CTRL. Statistically significant

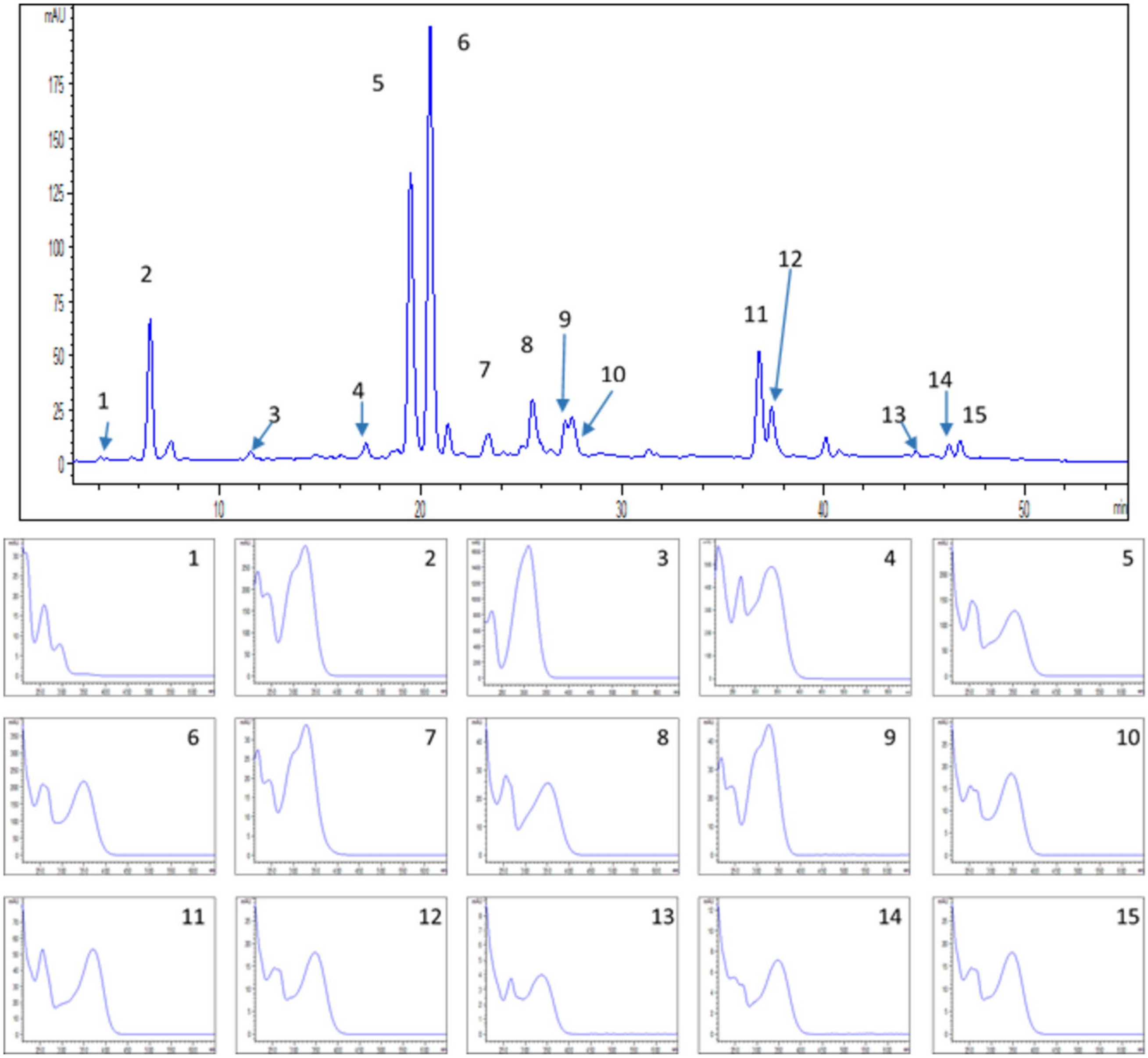


Fig. 1 HPLC finger print of G. verum extract. The numbers assigned to identified peaks referred to the following compounds (UV spectra of the identified compounds were presented in order as they are mentioned in this legend): 1. Protocatechuic acid; 2. Chlorogenic acid; 3. p-Coumaric acid; 4. Vitexin; 5. Rutin; 6. Luteolin-7-O-glucoside; 7. 3,5-di-O-caffeoylquinic acid; 8. Quercitrin; 9. Rosmarinic acid; 10. Diosmin; 11. Quercetin; 12. Luteolin; 13. Apigenin; 14. Chrysoeriol; 15. Isorhamnetin

increase in the GV and PSORI groups was observed at perfusion pressure starting from 60 to 120 cm H_2O compared to CTRL. The largest increase in CF was observed in PSORI + GV group on perfusion pressure starting from 80 to 120 cm H_2O compared to control values (Fig. 3f).

Oxidative stress

The measured values of oxidative stress (TBARS, NO_2-,O_2-, H_2O_2) were significantly increased in all coronary perfusion pressure in the PSORI group compared to the CTRL and GV groups. However, in the PSORI + GV group, it was observed significant drop in the measured parameters of oxidative stress compared to the PSORI group. In the GV group, values of TBARS and NO_2- were lower compared to the CTRL group (Fig. 4a–d).

Morphometric parameters of the left ventricle

The longitudinal section diameter of the cardiomyocytes was increased in the PSORI group by 45% and in the PSORI + GV by 21% compared to control values. By

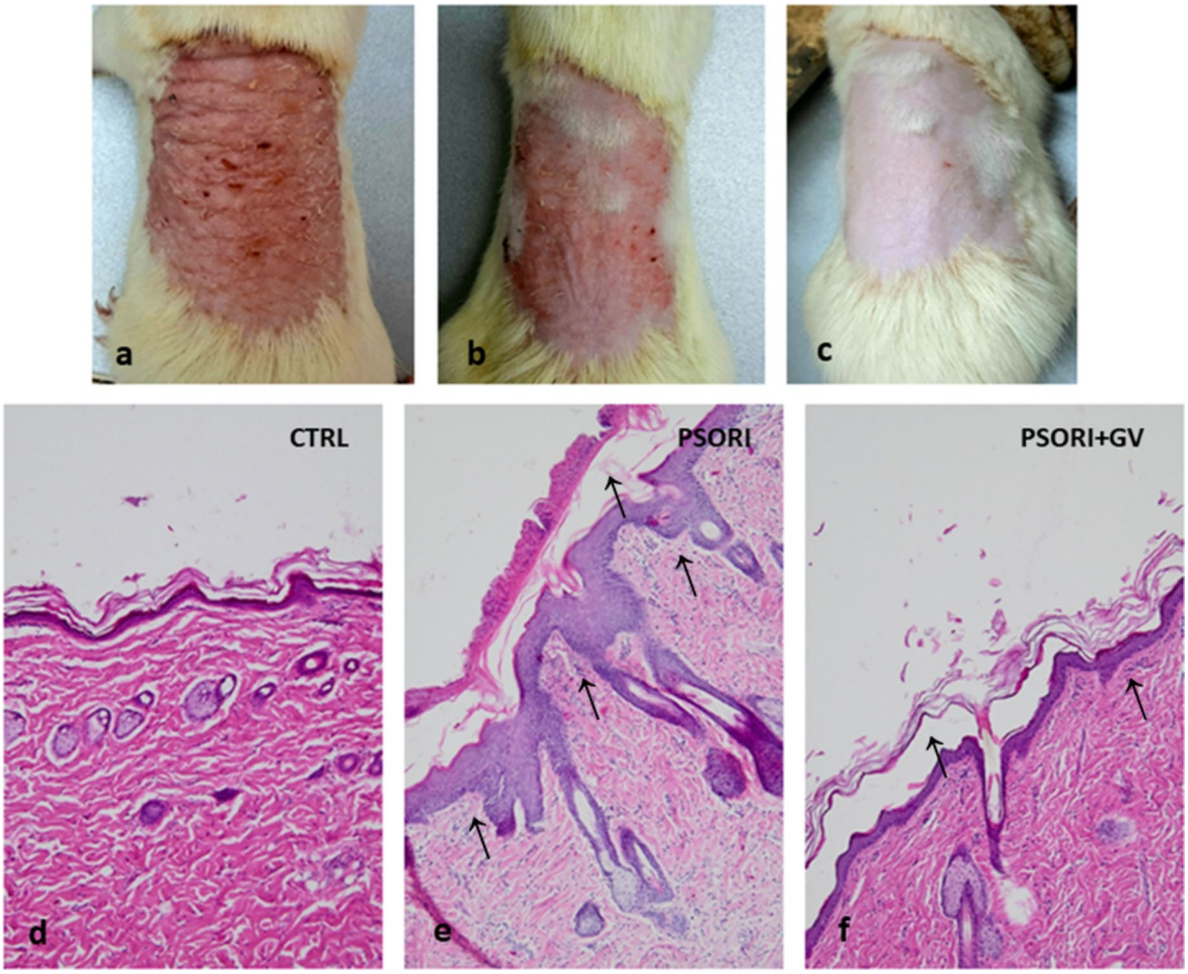


Fig. 2 Macroscopic (first row) and microscopic (second row) photos of psoriasis skin changes in rats: rat skin after induction of psoriasis (**a**), rat skin after 7 days of treatment with *G. verum* extract (**b**), rat skin after 14 days of treatment with *G. verum* extract (**c**), rats skin from control group (**d**), rat skin from psoriasis group (**e**), rat skin after 28 days of treatment with *G. verum* extract (**f**)

Table 2 Average values of body weight (BW), heart weight (HW), and index hypertrophy of heart (HW/BW ratio)

Groups	BW	HW	HW/BW ratio
CTRL	373.03 ± 3.22	1.29 ± 0.01	0.35 ± 0.01
GV	370.67 ± 8.12	1.28 ± 0.06	0.34 ± 0.01
PSORI	436.33 ± 15.324*#	1.72 ± 0.04*#	0.40 ± 0.01*#$
PSORI + GV	423.67 ± 5.09*#$	1.59 ± 0.02*#$	0.37 ± 0.01*#$

Results presented as mean value ± SD (n = 6). Statistical significance between groups is shown as *, #, $, * denotes vs. CTRL; # denotes vs. GV; $ denotes vs. PSORI; *CTRL* control, *GV* Galium verum, *PSORI* psoriasis

comparison between experimental groups, treatment with *G. verum* in psoriatic rats, significantly decreased the diameter of cardiomyocytes by 16% compared to psoriasis alone (Fig. 5a). Cross-section area of cardiomyocytes was increased by 47% in the PSORI group and by 24% in the PSORI + GV group compared to control. Administration of *G. verum* extract in psoriasis rats decreased cross-section area of cardiomyocytes by 16% compared to the PSORI group (Fig. 5b). Collagen content in the cardiac muscle was a significant increase in the PSORI group by 283% and in the PSORI + GV group by 188%, whereas in the GV group collagen was decreased by 13% compared to control values. By comparison between experimental groups, in the PSORI + GV group was observed decrease in collagen content by 34% compared to PSORI group (Fig. 5c). Microphotographs of H/E staining of the cardiac muscle tissue (Fig. 4, left row) in the CTRL and GV groups showed regular morphological structure without inflammation and hypertrophy of cardiomyocytes. In the PSORI group, it was observed irregular morphological structure with inflammation, hypertrophy of cardiomyocytes and increased interstitum space. However, treatment with *G. verum* extract in psoriatic rats visibly reduced morphological changes in relation to

Table 3 Average values of echocardiographic evaluation: IVSd - intraventricular septal wall thickness at end-diastole; LVIDd - left ventricle internal dimension at end-diastole; LVPWd - left ventricle posterior wall thickness at end-diastole; IVSs - intraventricular septal wall thickness at end-systole; LVIDs - left ventricle internal dimension at end-systole; LVPWs - left ventricle posterior wall thickness at end-systole; FS - fractional shortening percentage

	CTRL	GV	PSORI	PSORI + GV
IVSd (cm)	0.130 ± 0.006	0.127 ± 0.007	$0.143 \pm 0.007^{*\#}$	0.132 ± 0.007
LVIDd (cm)	0.885 ± 0.009	$0.830 \pm 0{,}010^{*}$	$0.928 \pm 0.022^{*\#}$	$0.841 \pm 0.018^{*}$
LVPWd (cm)	0.157 ± 0.005	0.148 ± 0.018	$0.182 \pm 0.033^{*\#}$	$0.177 \pm 0.007^{*\#}$
IVSs (cm)	0.190 ± 0.012	0.178 ± 0.006	$0.212 \pm 0.014^{*\#}$	$0.183 \pm 0.006^{\$}$
LVIDs (cm)	0.557 ± 0.009	0.525 ± 0.002	$0.642 \pm 0.044^{*\#}$	$0.540 \pm 0.021^{\$}$
LVPWs (cm)	0.250 ± 0.030	0.212 ± 0.024	$0.239 \pm 0.035^{\#}$	$0.295 \pm 0.011^{*\$}$
FS (%)	35.4%	36.75%*	$30.54\%^{*\#}$	$36.07\%^{\$}$

Results presented as mean value ± SD (n = 6). Statistical significance between groups is shown as *, #, $,: * denotes vs. CTRL; # denotes vs. GV; $ denotes vs. PSORI; *CTRL* control, *GV* Galium verum, *PSORI* psoriasis

psoriasis. In the PSORI + GV group, it was observed regular morphological structure without inflammation, but with mild signs of hypertrophy of cardiomyocytes. Microphotographs of Picro Sirius red staining of the cardiac muscle tissue presented in Fig. 6 (right row) shows a red deposits of collagen fibers. In the PSORI group the coarser and more bundled collagen fibers/deposits were observed. Administration of *G. verum* to psoriatic rats, lowered the overall level of collagen in the heart, which is also shown by image analysis.

Discussion

Psoriasis represents a chronic inflammatory skin disease that has recently been considered a systemic entity. In recent years, it has been increasingly reported that the disease itself causes harmful effects on the cardiovascular system and that the effects of the disease depend on the degree of skin changes. However, there is lack of data to confirm this fact. Knowing about the numerous benefits of phytotherapy on various organ systems, our guiding idea for this research was exactly that, to show that the administration of plant extracts can alleviate the harmful effects of diseases on the heart and at the same time improve local changes in the skin. Based on *G. verum* application in traditional medicine in the topical treatment of psoriasis, as well as on previously published studies of its cardioprotective effects on ischemia–reperfusion injury of the isolated heart of normotensive and spontaneously hypertensive rats, *G. verum* extract was chosen for further investigation. According to our knowledge, this was the first study that investigated cardioprotective effects of systemic administration of *G. verum* extract on function and morphology of the isolated heart of psoriatic rats.

Galium sp. have a long history of use as a traditional medicinal plant. Recently, considerable research on this plant was performed, revealing that this plant might be considered effective, safe, and easily accessible [23]. Chemical analysis of ethanolic extract in our research revealed the presence of various phenolic compounds, while rutin and chlorogenic were the most abundant. Previously confirmed promising antioxidant, anti-inflammatory and anti-proliferative actions of these compounds [2] encouraged us to comprehensively assess future application of G. verum ethanolic extract in psoriasis management in rats.

In our study, firstly, we demonstrated that systemic administration of *G. verum* extract improved local changes in the skin of psoriatic rats. By determining the PASI score after the induction of psoriasis and after 7 and 14 days of treatment, a significant reduction in erythema and desquamation was observed. Unfortunately, we do not have the macroscopic photos after 21 and 28 days of treatment, because the hair has grown and covered the skin, but we have histological confirmation of the effectiveness of the *G. verum* extract in the treatment of psoriatic skin of rats. In addition, by histological confirmation, we demonstrated that the administration of *G. verum* extract reduced inflammatory changes on the skin, which represented direct confirmation of anti-inflammatory effects of *G. verum* extract administration. This was in accordance with the literature data that revealed the *G. verum* anti-inflammatory potential in the treatment of aphthous stomatitis [19].

An echocardiographic evaluation of heart function showed that psoriasis increased IVSd, LVIDd, LVPWd, IVSs and LVIDs, whereas decrease of FS was registered. This result indicated that the disease itself has a negative influence on cardiac function by inducing hypertrophy of the left heart. On the other hand, four week treatment with *G. verum* in psoriatic rats decreased IVSs, LVIDs, LVPWs and increased FS compared to psoriasis. These results confirmed that this plant reduced cardiac hypertrophy and improved cardiac function. Similar findings were reported by Bradic and coworkers [31]. They demonstrated that 4-week treatment of *G. verum* in spontaneously hypertensive rat decreased LVIDd, LVIDs, and LVPWd and increased FS

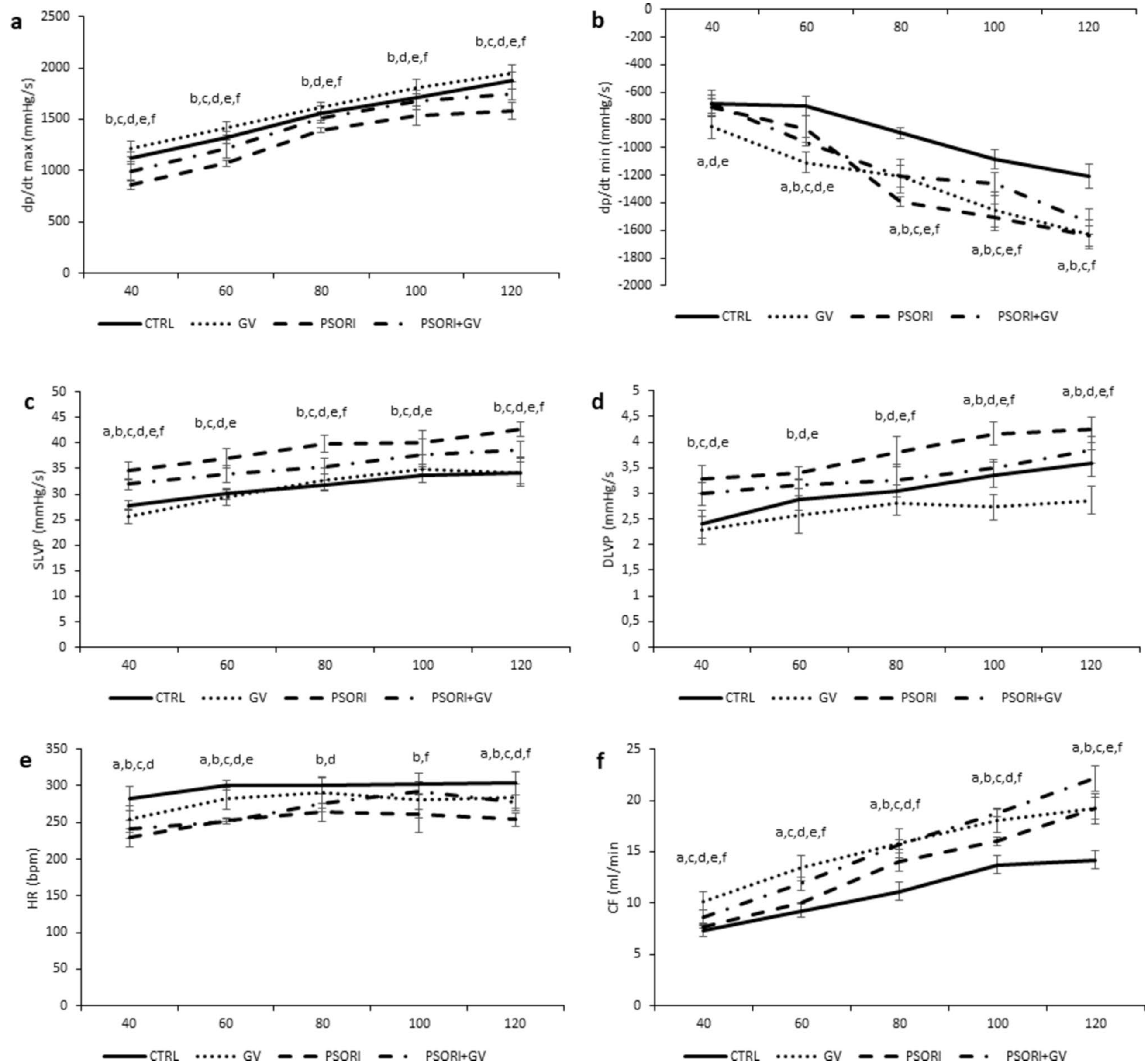


Fig. 3 Mean values of *dp/dt max* maximum rate of pressure development in the left ventricle (**a**), *dp/dr min* minimum rate of pressure development in the left ventricle (**b**), *SLVP* systolic left ventricular pressure (**c**), *DLVP* diastolic left ventricular pressure (**d**), *HR* heart rate (**e**), *CF* coronary flow (**f**). Each value represents the mean ± SD (n = 6). Statistical significance between groups is shown as a, b, c, d, e, f: a- denotes GV vs. CTRL; b- denotes PSORI vs. CTRL; c- denotes PSORI vs. GV; d- denotes GV vs. CTRL; e- denotes PSORI + GV vs. GV; f- denotes PSORI + GV vs. PSORI. *CTRL* control; *GV G. verum*; *PSORI* psoriasis; *PSORI + GV* psoriasis plus *G. verum*

[31]. Previous research demonstrated that natural compounds from a class of flavonoids might be responsible for reduction of endothelial dysfunction and cardiac hypertrophy in spontaneously hypertensive rats [31]. Furthermore, in models of left ventricular failure, the positive effect of isoflavones was demonstrated on overall cardiovascular markers, including increase of FS percentage [32]. Moreover, tannins, both hydrolyzable and condensed, have demonstrated cardioprotective effects in preclinical studies, showing potential to improve cardiac hypertrophy and attenuate different cardiac disorders. Therefore, we assumed that flavonoids along with tannins from *G. verum* might be responsible for reducing left heart hypertrophy and increasing FS in the PSORI + GV group.

Present study demonstrated that disease itself caused several changes in the heart which were reflected in impaired function and morphology of the heart. By measuring dp/dt max and dp/dt min we estimated cardiac contractility and

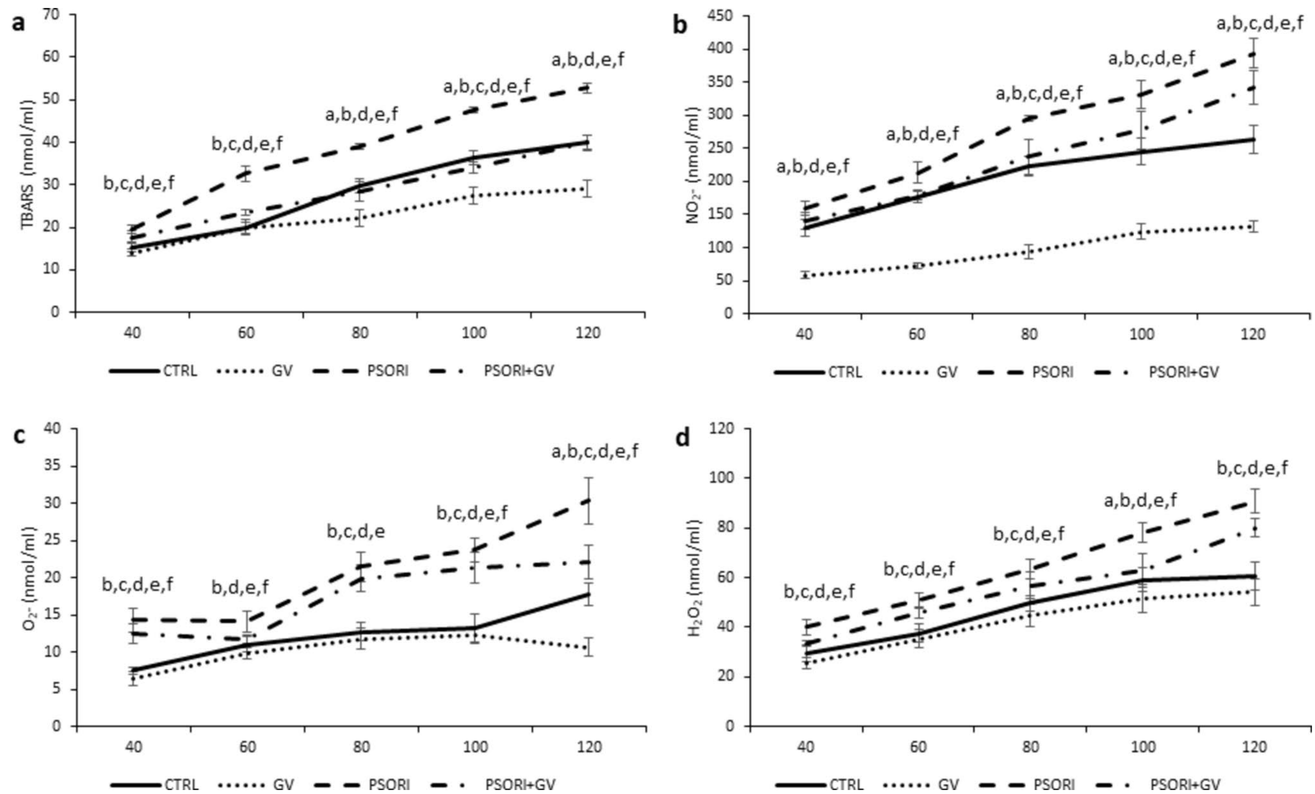


Fig. 4 Mean values of TBARS (a), NO_{2-} (b), O_{2-} (c), H_2O_2 (d). Each value represents the mean ± SD (n = 6). Statistical significance between groups is shown as a, b, c, d, e, f: a- denotes GV vs. CTRL; b- denotes PSORI vs. CTRL; c- denotes PSORI vs. GV; d- denotes GV vs. CTRL; e- denotes PSORI + GV vs. GV; f- denotes PSORI + GV vs. PSORI: *CTRL* control; *GV G.verum*; *PSORI* psoriasis; *PSORI + GV* psoriasis plus *G.verum*

relaxation. It was observed that the disease itself, reduced cardiac contractility (inotropic properties) at all perfusion pressures. The values of cardiac relaxation (lusitropic properties) were reduced (more negative values) on higher perfusion pressures. On the other hand, treatment with *G. verum* nullifies negative effects of disease by improving cardiac contractility and relaxation especially on higher coronary perfusion pressures. Similar positive effects of *G. verum* were estimated in studies of Bradic and colleges confirming that administration of *G. verum* extract improved inotropic and lusitropic properties of the heart in doxorubicin treated rats [20], as well as in ischemic/reperfusion injury of the heart [17].

Literature data suggest a link between psoriasis and hypertension, but the exact mechanism, which is involved in this, is still unknown. The role of angiotensin II and oxidative stress should not be neglected. Namely, angiotensin II may promote T-cell proliferation, inflammation and development of arteriosclerosis whereas increased levels of reactive oxygen species may harm vasodilation that is dependent on the endothelium [33]. The present study revealed that the values of systolic and diastolic left ventricle pressure were increased in the PSORI group on all perfusion pressure. The increase in values of pressure may lead to hypertrophy of the heart. Bearing in mind that the largest increase in SLVP and DLVP was in the PSORI group, we assumed that one of the possible mechanism for this might be the hypertrophy of the heart, which was confirmed in our study by echocardiographic examination and morphometric analysis of cardiac muscle cells. The PSORI group had a reduced rate of cardiac relaxation, which led to a rise in DLVP levels. In this instance, the left ventricle received less blood during diastole, and any filling deficit could affect systolic function. The precise mechanism via which psoriasis influences this process has still remained undefined. We can only speculate that heart hypertrophy, particularly given the psoriasis rats' hearts higher collagen content, may be one of the factors impairing the hearts ability to relax. Also, the increase in oxidative stress in the PSORI group might be considered as the mechanism responsible for this result. In psoriasis patients, adipose tissue represents the main source of angiotensinogen. By conversion of angiotensinogen, angiotensin II is formed [33]. Bearing in mind that the largest increase in BW was observed in the PSORI group, the logical assumption might be that there was also an increase in angiotensin II. This finding might put further light in explanation of

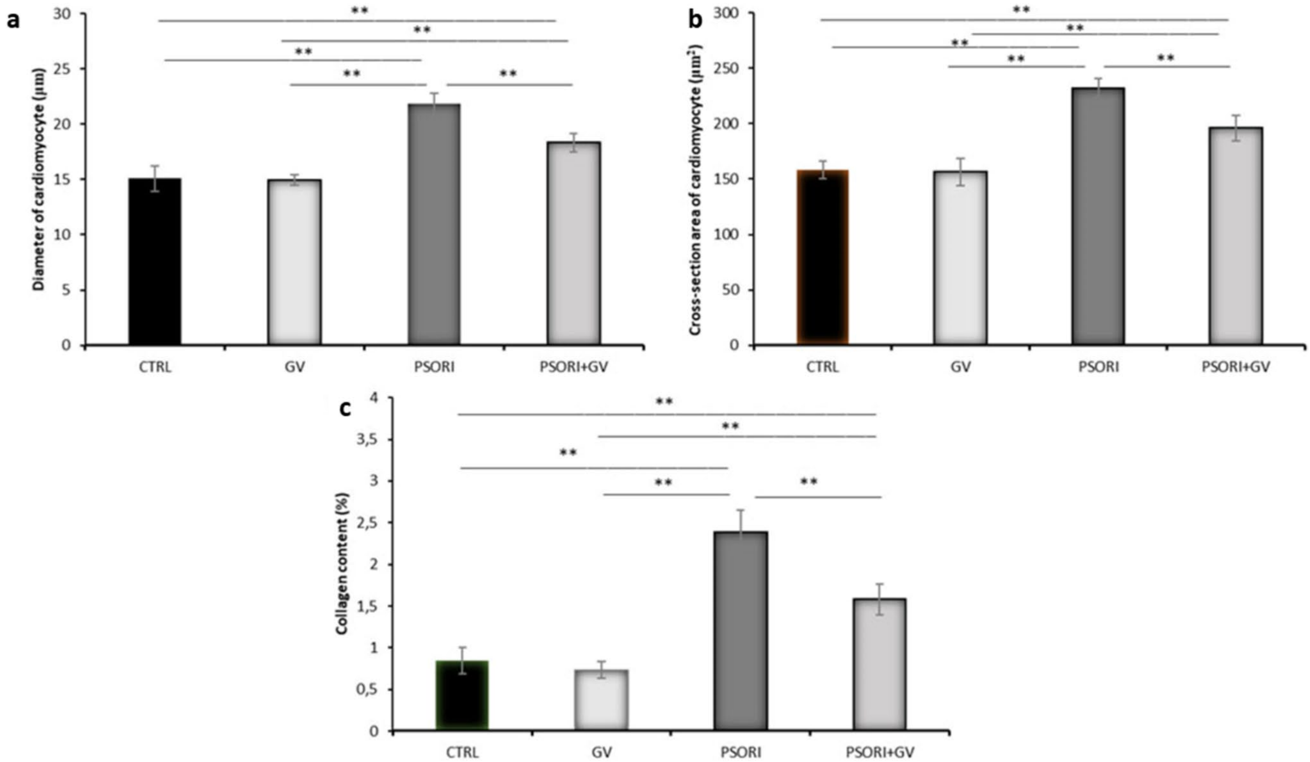


Fig. 5 Diameter of the cardiomyocytes (**A**), cross-section area of the cardiomyocytes (**B**) and collagen content (**C**) after second, third and fourth experimental week. Each value represents the mean ± SD (n = 6)

obtained results. On the other hand, treatment with *G. verum* in psoriatic rats improved SLVP and DLVP values. This result were in accordance with the findings that *G. verum* extract protected the hearts systolic and diastolic functions, which was confirmed in a previous study [20].

The reduction of heart rate was noticed in the PSORI group, which could be a consequence of autonomic nervous activity of the heart. On the other hand, administration of *G. verum* extract improved heart rate especially at higher coronary perfusion pressure in psoriatic rats. The results obtained in this study were direct confirmation of the cardioprotective effect of *G. verum* extract on the cardiac function of psoriatic rats. Similar findings were obtained in the I/R injury model and might suggest that polyphenols in *G. verum* mostly contribute to triggering cardioprotection [17]. In line with the previous studies [34, 35], our research data highlight protective impact of extract rich in rutin and chlorogenic acid on heart function and supports the evidence for valuable action of these compounds in prevention of cardiac dysfunction.

It is well known that imbalance of the redox state leads to oxidative damage to various cellular components. A disturbed balance between ROS (reactive oxygen species) and RNS (reactive nitrogen species) causes oxidative stress, which was confirmed in our study. Namely, the increased values of TBARS, NO_2-, O_2- and H_2O_2 in the coronary venous effluent in the PSORI group represented a direct indicator of increased oxidative stress in the heart itself. On the other hand, the administration of *G. verum* to psoriatic rats significantly reduced the values of all measured parameters of oxidative stress, confirming the strong antioxidant effect of this plant. This was in line with the results presented in the study by Bradic and colleagues, who showed that the administration of *G. verum* extract reduced the oxidative stress in a model of ischemia–reperfusion injury to the heart [17]. What is very important, the conductaed experiments in this study demonstrated that the administration of *G. verum* extract in healthy animals significantly lowered the values of TBARS and NO_2- in comparison to the control values, which confirmed the antioxidant effect of this plant.

The existence of a possible overlap of immunological pathways in psoriasis and cardiovascular diseases [36] was confirmed in this study. By morphological examination of heart tissue, we observed an increased presence of inflammatory cells in the group of animals with psoriasis. Opposite of this, treatment with *G. verum* extract in psoriatic rats reduced inflammation in the heart, which was direct confirmation of anti-inflammatory effects of this plant.

That disease itself induces hypertrophy of the heart was confirmed by determining the index of hypertrophy of the

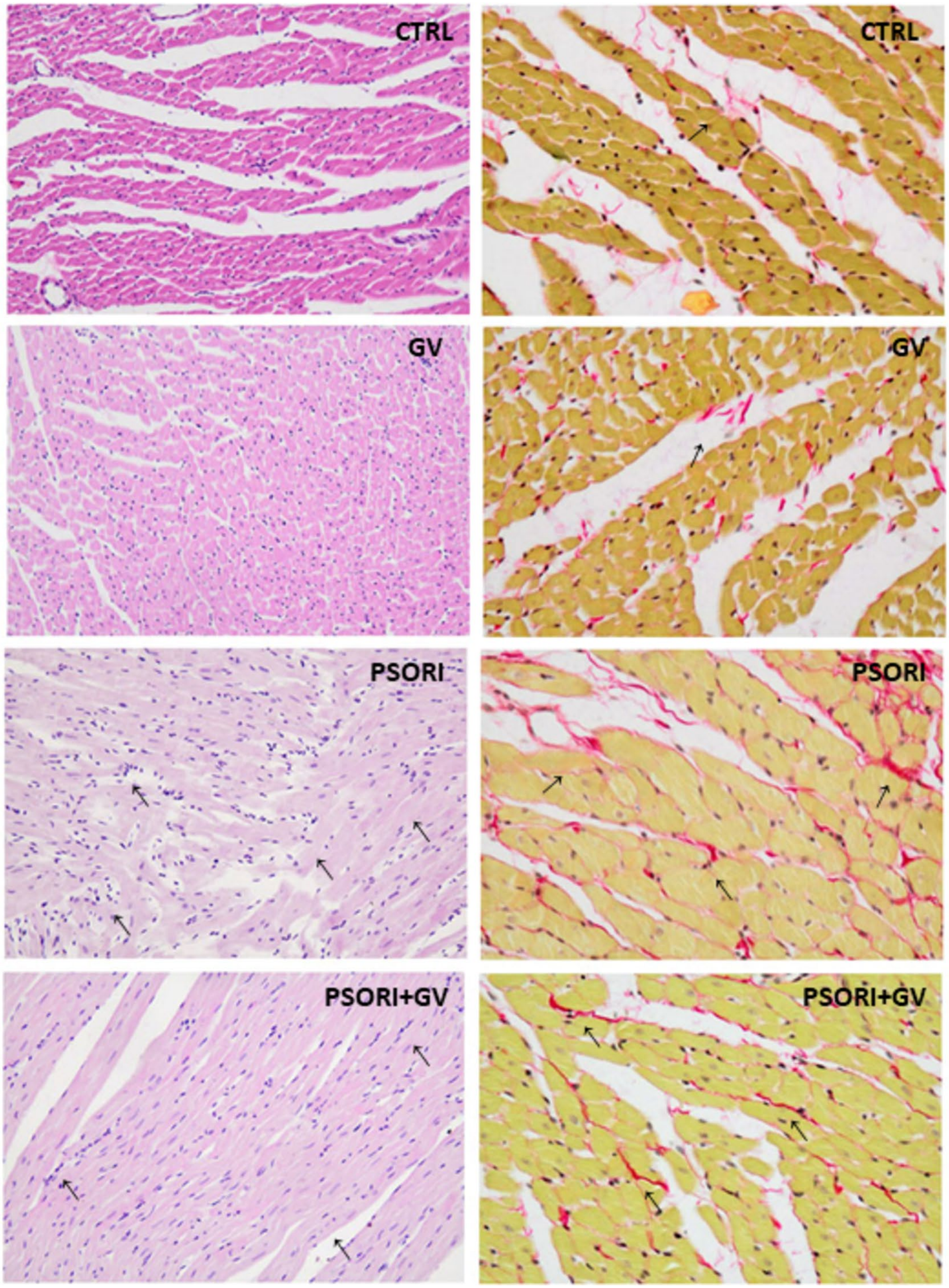


Fig. 6 Representative microphotographs of the heart staining with H/E (left row) and Picro-sirius red staining (right row). Magnification 200x. Arrows indicate sites of histological changes, such as cardiomyocyte hypertrophy, inflammation, and red-stained collagen fibers

heart, by echocardiographic examination and by measuring the diameter and cross-section area of cardiomyocytes. By morphometric analysis of cardiomyocytes, we showed that psoriasis significantly increased the size of cardiomyocytes (diameter and cross-section area). Additionally, the values of cardiac hypertrophy index (HW/BW ratio) in rats with psoriasis were correlated with both cardiomyocyte hypertrophy and ehocardiographic evaluation. Based on the achieved results, it was revealed that the disease itself affected the development of cardiac fibrosis. Abovementioned findings together implied that the disease impairs the function of the heart. On the other hand, administration of *G. verum* extract in rats with psoriasis significantly reduced cardiomyocyte hypertrophy and collagen content in the heart. We assumed that rutin, and chlorogenic acid as the most abundant polyphenols, in *G. verum* extract, were responsible for the reduction of cardiomyocyte hypertrophy in psoriatic rats [31] as well as for the reduction of collagen content in the heart.

The novelty of this study is that, for the first time, the effects of systemic application of *G. verum* extract on an animal model of psoriasis were demonstrated. In addition to the cardioprotective effects, the application of the extract also improved local changes in the skin of rats with psoriasis. Given that psoriasis is considered a systemic entity that affects various organ systems, systemic application of *G. verum* extract could significantly alleviate the effects of the disease. The clinical significance of this study is manifold. Bearing in mind that the use of conventional medicine resulted in numerous side effects, the inclusion of plants, as an independent or combined therapy, would significantly reduce the systemic negative effect of the disease whereas at the same time improving the quality of life of patients suffering from psoriasis. Combining synthetic drugs and herbal medicine could reduce the dose of synthetic drugs, and therefore the cost of treatment. After confirming the effectiveness of plant extracts in an animal model of the disease, future research could be translated to humans in order to confirm the effectiveness of the application of the natural compound in the treatment of psoriasis as one of the modalities of adjuvant therapy. In addition, the effects of concomitant application of synthetic and herbal medicines in the treatment of psoriasis should be investigated.

Conclusion

Generally, we can conclude that disease itself causes harmful effects on the heart by decreasing cardiac contractility and relaxation and increasing systolic and diastolic left ventricle pressure, oxidative stress biomarkers, hypertrophy of cardiomyocytes and cardiac fibrosis. However, treatment with *G. verum* in psoriatic rats improved cardiac function, oxidative stress biomarkers, decreases hypertrophy of cardiomyocytes and collagen content in the heart as well as local changes on psoriatic skin. Further investigation is needed to examine the potential therapeutic effects of *Galium verum* extract in the treatment of psoriasis.

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Author contributions M. C.: conceptualization, methodology, investigation, formal analysis, writing -original draft preparation, writing-review and editing. V. J.: resources, validation, supervision, project administration. M. N.: methodology, investigation. N. J.: investigation, supervision. J. B.: extract preparation, investigation, supervision. J. N.: investigation, supervision. Aleksandar Kocovic: methodology, investigation. M. S.: methodology, investigation. V. T.: methodology, investigation. A. Z.: methodology, investigation. B. D.: methodology, investigation. M. K.: methodology, investigation. K. M.: methodology, investigation. J. S.: conceptualization, methodology, investigation, formal analysis, writing - review and editing, validation, supervision. All authors have read and agreed to the published version of the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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